



DATASHEET

JMS581DL **USB 3.2 Gen2 to SATA 6Gb/s & PCIe Gen3 x2** **Bridge Controller**

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Revision History

Revision number	Effective date	Description of revision		Author
		Reference	Description of change	
1.00	12/10/2022	-	Initial release.	George Chao
1.01	03/27/2023	Section 6.2 Section 6.6 Section 7.2	1. Update power dissipation 2. Update VCCK and AVDDL voltage 3. Update explanation of part number	George Chao
1.02	2026/04/28	-	Update JMicon company address	Nick Hung

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1 Overview

JMS581DL is a bridge controller between USB 3.2 Gen 2 10Gb/s to either SATA 6Gb/s or PCIe NVMe Gen 3x2. It combines the USB devices with both PCIe protocol and SATA protocol bridge controllers. The upstream port provides a USB 10Gb/s performance, and the downstream ports supports SATA 6Gb/s and PCIe NVMe Gen 3x2 protocol. It is able to compatible with SATA and PCIe NVMe storage device such as HDD, SSD and CFast, CFexpress memory card. Users can easily connect their storage devices, because this controller is able to operate in USB to PCIe mode, or USB to SATA mode and switch automatically.

Moreover, JMS581DL has USB Type-C™ connectivity built in to the controller that any device using JMS581DL can have a USB Type-C™ connector without adding any additional peripheral part. It can reduce printed circuit board area for the system designs.

JMS581DL supports TRIM to the NAND flash-based storages and enable transmit and receive data by both of the USB Mass Storage Class Bulk-Only Transport (BOT) and USB Attached SCSI Protocol (UASP) to and from the host respectively. The data storage devices can achieve its summit of performance by taking advantage of these built-in unmatched features.

JMS581DL is well equipped for power management that it can meet a wide variety of power requirements from different scales of data storage systems: external storage enclosure (M.2 combo enclosure), adapter and portable storage applications.

Owing to its USB Type-C™ connectivity, JMS581DL can work with some power management controllers to a USB Power Delivery (PD) enabled data storage device. The data storage devices having SSDs of large capacity can accept the electrical power from sources of energy, such as hosts acting as a power provider of USB PD to supply sufficient electricity to the device after they negotiate with each other, without plugging in.

2 Features

2.1 General Features

- Design for Windows 7, Windows 10 and MAC 10.10.5 or later version
- Provide 8 hardware controlled PWMs
- Provide software utilities for downloading the upgraded firmware code under USB 2.0/ USB 3.2 Gen1 and USB 3.2 Gen2
- 144TFBGA (9x9mm²) package
- Support 25MHz external crystal
- Support 3.3V I/O
- 32 GPIOs for customization

2.2 Universal Serial Bus

- Comply with USB 3.2 Gen 1 and Gen 2 Specification,
- Comply with USB Mass Storage Class, Bulk-Only Transport Specification (Revision 1.0)
- Comply with USB Attached SCSI Protocol (UASP) Specification (Revision 4)
- Integrate with USB Type-C™ multiplexer & configuration channel (CC) logic
- Support USB Super-Speed/ High-Speed/ Full-Speed Operation
- Support USB 2.0/ USB 3.2 Gen1/ Gen2 power saving mode
- Support external SPI NVRAM for Vendor VID/PID of USB 2.0/USB 3.2 Gen1/2 device controller

2.3 PCI Express

- Comply with PCI Express Base Specification Revision 3.1a
- Comply with NVM Express 1.3
- Support TRIM to the SSD

2.4 Serial ATA

- Comply with SATA Specification (Revision 3.1)
- Support TRIM to SATA
- Support Native Command Queue (NCQ)

3 Block Diagram

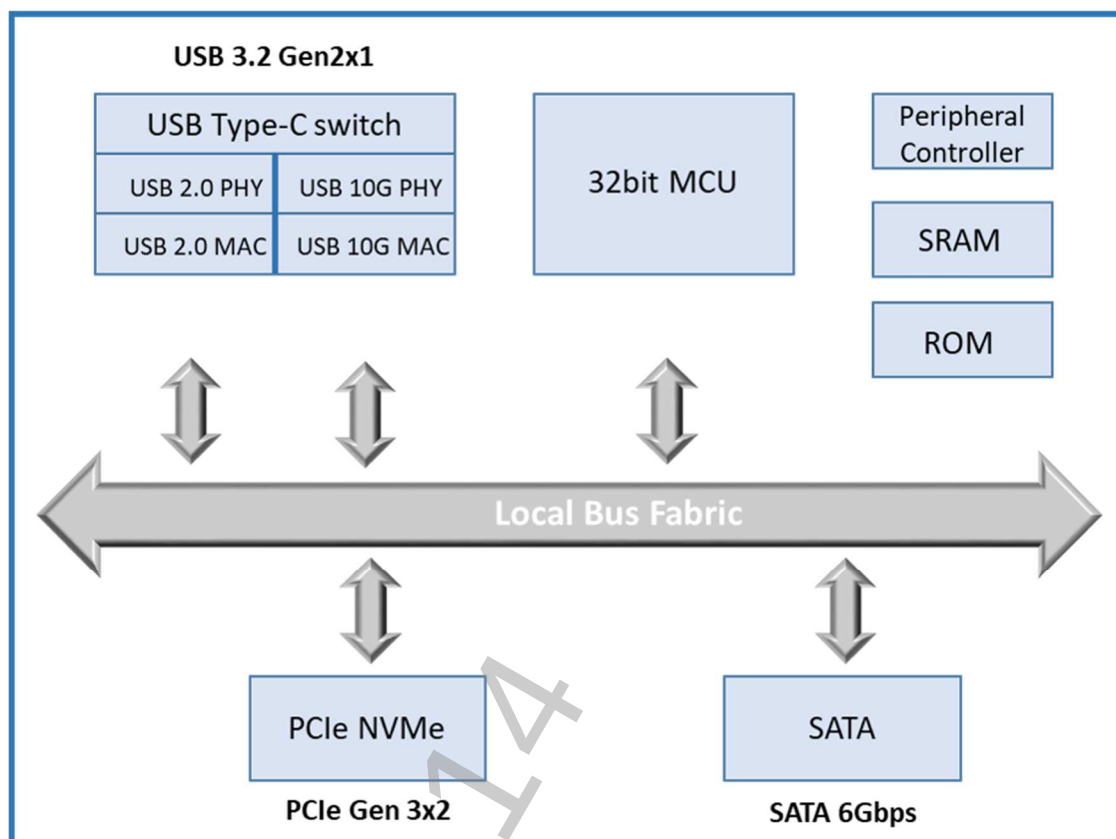


Figure 1 Block Diagram

4 Application

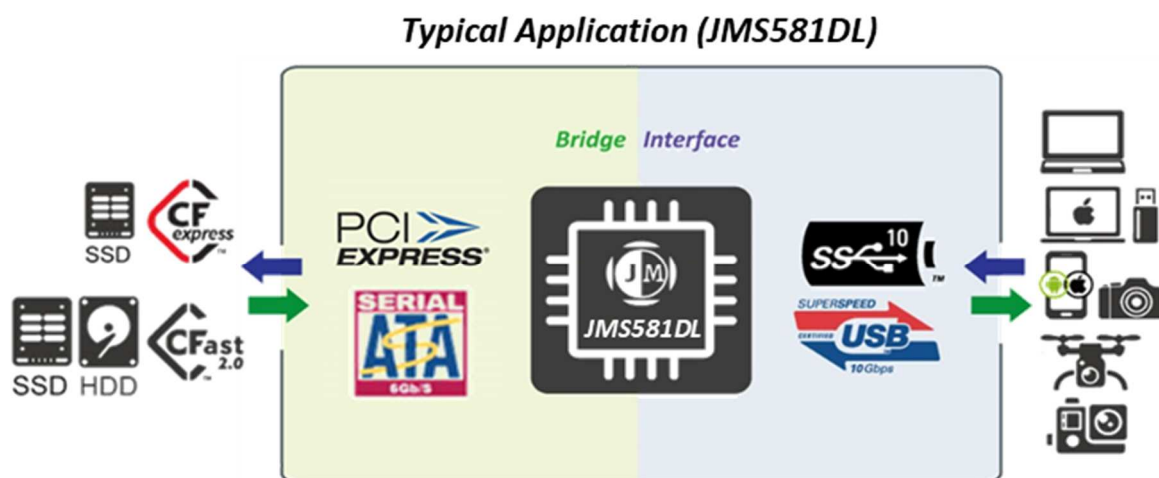


Figure 2 Application Scenarios – Dual Protocol

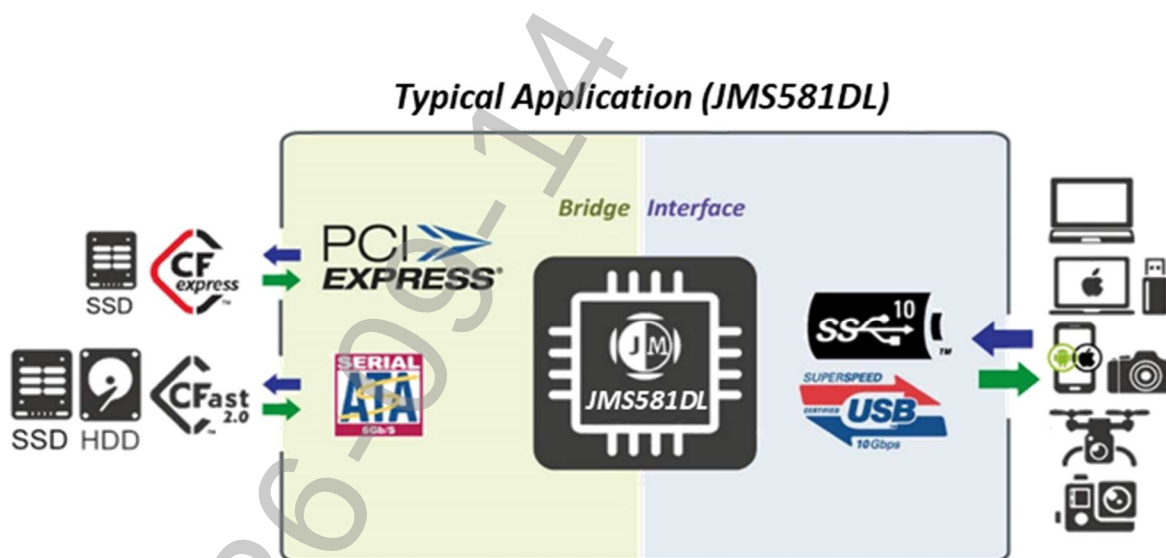


Figure 3 Application Scenarios – 2 LUNs

5 Package and Ball Assignments

5.1 Ball Diagram

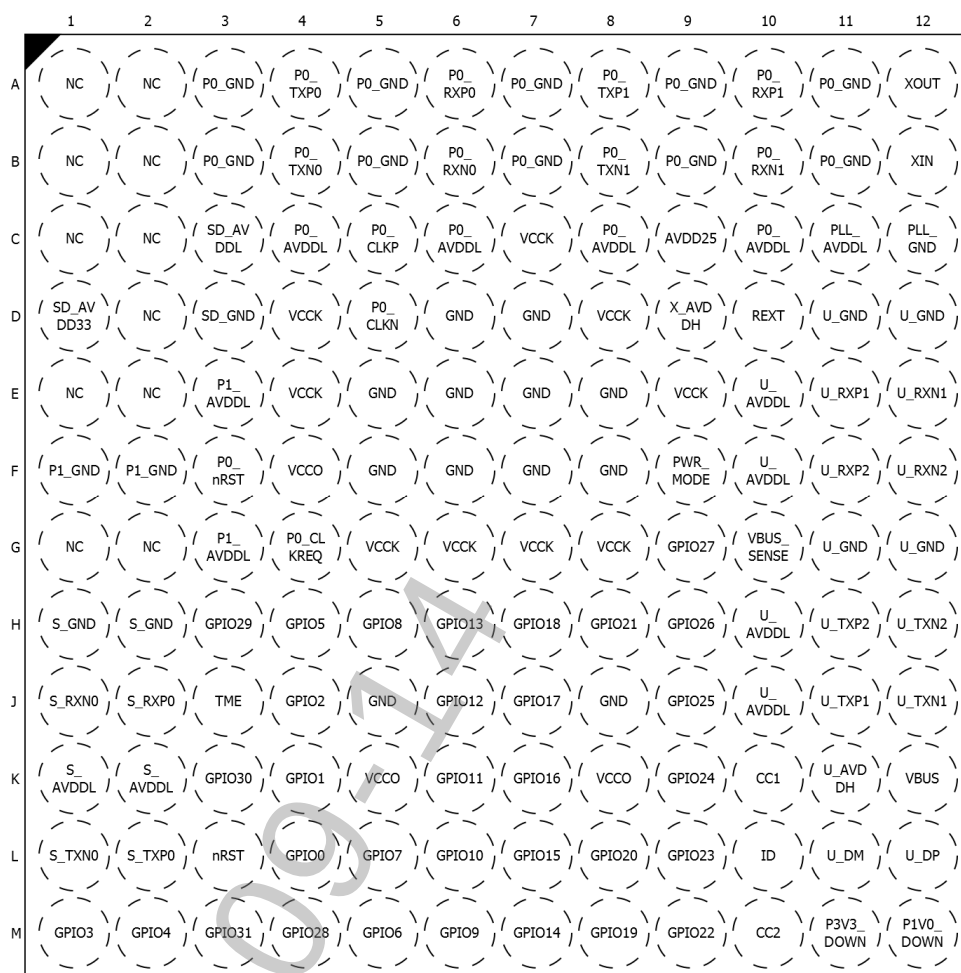


Figure 4 Ball Assignment of JMS581DL

5.2 Mechanical Dimensions

Symbol	Dimension in mm			Dimension in inch		
	MIN	NOM	MAX	MIN	NOM	MAX
A	1.03	1.10	1.17	0.041	0.043	0.046
A1	0.16	0.21	0.26	0.006	0.008	0.010
A2	0.84	0.89	0.94	0.033	0.035	0.037
c	0.32	0.36	0.40	0.013	0.014	0.016
D	8.90	9.00	9.10	0.350	0.354	0.358
E	8.90	9.00	9.10	0.350	0.354	0.358
D1	---	8.25	---	---	0.325	---
E1	---	8.25	---	---	0.325	---
e	---	0.75	---	---	0.030	---
b	0.25	0.30	0.35	0.010	0.012	0.014
aaa	0.15			0.006		
ccc	0.10			0.004		
ddd	0.08			0.003		
eee	0.15			0.006		
fff	0.08			0.003		
MD/ME	12/12					

NOTE :

1. CONTROLLING DIMENSION : MILLIMETER.

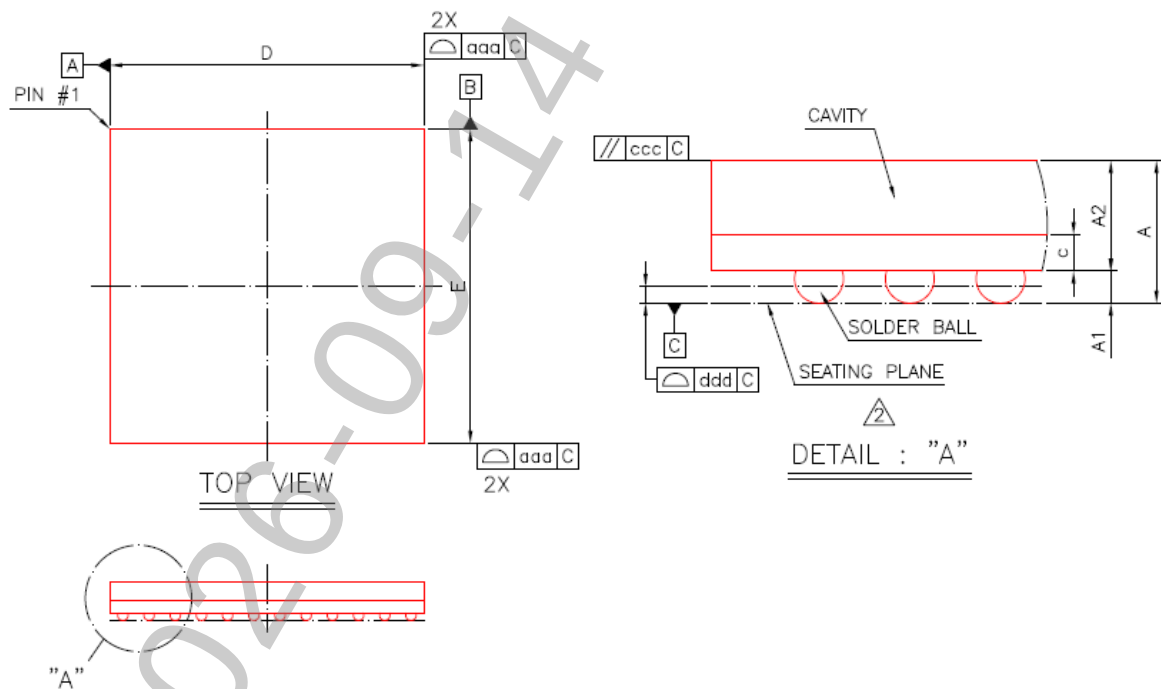
② PRIMARY DATUM C AND SEATING PLANE ARE DEFINED BY THE SPHERICAL CROWNS OF THE SOLDER BALLS.

③ DIMENSION b IS MEASURED AT THE MAXIMUM SOLDER BALL DIAMETER, PARALLEL TO PRIMARY DATUM C.

4. SPECIAL CHARACTERISTICS C CLASS: ccc,ddd(SPIL STANDARD)

④ THE PATTERN OF PIN 1 FIDUCIAL IS FOR REFERENCE ONLY.

6. REFERENCE DOCUMENT : JEDEC PUBLICATION 95
DESIGN GUIDE 4.5



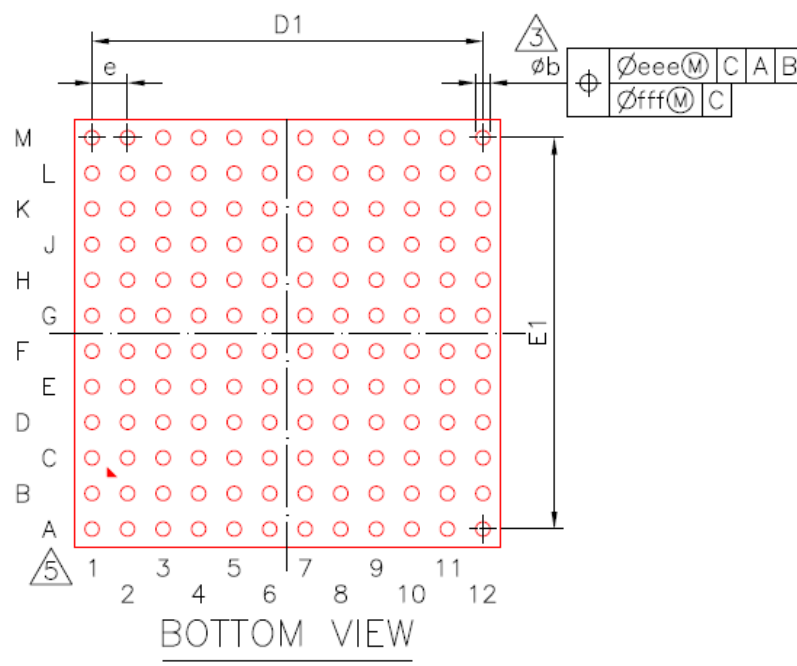


Figure 5 Package Outline Drawing of 144TFBGA 9x9mm²

5.3 Signal Descriptions

5.3.1 Pin Type Definitions

Table 1 Pin Type Definitions

Pin Type	Definitions
A	Analog
D	Digital
P	Power
I	Input
O	Output
IO	Bi-directional
OD	Open-Drain
H	Internal weak pull-high
L	Internal weak pull-low

5.3.2 Ball Descriptions

PCIe Interface Signals

Table 2 PCIe Interface Signals

Signal Name	Pin Number	Type	Description
P0_RXN1	B10	AI	PCIe Port0 RX- Signal of Lane 1
P0_RXP1	A10	AI	PCIe Port0 RX+ Signal of Lane 1
P0_TXN1	B8	AO	PCIe Port0 TX- Signal of Lane 1 A 220 nF capacitor should be connected between this pin and PCIe connector.
P0_TXP1	A8	AO	PCIe Port0 TX+ Signal of Lane 1 A 220 nF capacitor should be connected between this pin and PCIe connector.
P0_RXN0	B6	AI	PCIe Port0 RX- Signal of Lane 0
P0_RXP0	A6	AI	PCIe Port0 RX+ Signal of Lane 0
P0_TXN0	B4	AO	PCIe Port0 TX- Signal of Lane 0 A 220 nF capacitor should be connected between this pin and PCIe connector.
P0_TXP0	A4	AO	PCIe Port0 TX+ Signal of Lane 0 A 220 nF capacitor should be connected between this pin and PCIe connector.
P0_CLKP	C5	DO	PCIe Port0 Reference Clock P 100MHz reference clock for Device.
P0_CLKN	D5	DO	PCIe Port0 Reference Clock N 100MHz reference clock for Device.
P0_CLKREQ	G4	DIO	PCIe Port0 This is for L1 substrate
P0_nRST	F3	DO	PCIe Port0 Reset for Device

USB 3.2 Gen2 Interface Signals

Table 3 USB 3.2 Gen2 Interface Signals

Signal Name	Pin Number	Type	Description
U_RXP2	F11	AI	Super Speed RX+ 2 signal
U_RXN2	F12	AI	Super Speed RX- 2 signal
U_RXN1	E12	AI	Super Speed RX- 1 signal
U_RXP1	E11	AI	Super Speed RX+ 1 signal
U_TXP2	H11	AO	Super Speed TX+ 2 signal A 100 nF capacitor should be connected between this pin and USB connector.
U_TXN2	H12	AO	Super Speed TX- 2 signal A 100 nF capacitor should be connected between this pin and USB connector.
U_TXN1	J12	AO	Super Speed TX- 1 signal A 100 nF capacitor should be connected between this pin and USB connector.
U_TXP1	J11	AO	Super Speed TX+ 1 signal A 100 nF capacitor should be connected between this pin and USB connector.
REXT	D10	AI	External Reference Resistance A 12 kΩ±1% external resistor should be connected to this pin.

USB 2.0 Interface Signals

Table 4 USB 2.0 Interface Signals

Signal Name	Pin Number	Type	Description
U_DP	L12	AIO	USB 2.0 Bus D+ Signal
U_DM	L11	AIO	USB 2.0 Bus D- Signal
ID	L10	AI	OTG ID Input 0: It is connected to ground. USB is as a host. Otherwise: USB is as a device.
VBUS	K12	PI	USB 5V VBUS power for OTG VBUS voltage detection

Signal Name	Pin Number	Type	Description
VBUS_SENSE	G10	DI	VBUS Sense Input It is used for VBUS detection. There is a voltage divider circuit on a board. Power source is connected from VBUS 5V. The output is connected to the signal for VBUS detection.

USB Type-C™ Configuration Channel

Table 5 USB Type-C™ Configuration Channel

Signal Name	Pin Number	Type	Description
CC1	K10	AI	CC Pin1 input for voltage detection The maximum tolerant input voltage is 3.3V.
CC2	M10	AI	CC Pin2 input for voltage detection The maximum tolerant input voltage is 3.3V.

Serial ATA Interface Signals

Table 6 Serial ATA Interface Signals

Signal Name	Pin Number	Type	Description
S_RXP0	J2	AI	SATA Port RX+ Signal A 10 nF capacitor should be placed between this pin and SATA connector.
S_RXN0	J1	AI	SATA Port RX- Signal A 10 nF capacitor should be placed between this pin and SATA connector.
S_TXN0	L1	AO	SATA Port TX- Signal A 10 nF capacitor should be placed between this pin and SATA connector.
S_TXP0	L2	AO	SATA Port TX+ Signal A 10 nF capacitor should be placed between this pin and SATA connector.

Crystal Interface

Table 7 Crystal Interface

Signal Name	Pin Number	Type	Description
XIN	B12	AI	Crystal Input/ Oscillator Input It is connected to a 25MHz crystal or crystal oscillator. The variation range should be $\pm 30\text{ppm}$. And the input voltage should range in $3.3\text{V} \pm 5\%$.

XOUT	A12	AO	Crystal Output It is connected to a crystal. While crystal oscillator is applied, this pin should be reserved for No Connection (NC). The output variation range is around $\pm 30\text{ppm}$ (input dependent). And the input voltage range is $3.3\text{V} \pm 5\%$ (input dependent).
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Control and GPIO Interface

Table 8 Control and GPIO Interface

Signal Name	Pin Number	Type	Description
nRST	L3	DI	System Global Reset Input Active-low to reset the entire chip. An external RC should be connected to this pin.
TME	J3	DI	MP Test Mode Enable The pin is reserved for IC mass production testing. Keep this pin to logic "0" in normal operation.
PWR_MODE	F9	DI	Power Domain Control 0: Only ROM, ILM, DLM and always-on power domains are turned on during power-on 1: All power domains are turned on during power-on.
P3V3_DOWN	M11	DO	Power 3.3V Voltage Control 0: low voltage, 1: normal voltage during power on.
P1V0_DOWN	M12	DO	Power 1.0V Voltage Control 0: low voltage, 1: normal voltage during power on.
GPIO [0]	L4	DIOH	Serial Flash (SO) After power on status detecting, this pin becomes Data Output of serial flash. This pin is by default set to input.
GPIO [1]	K4	DIOH	Serial Flash (SCK) This pin is Serial Flash Data Clock (SCK) of serial flash. This pin is by default set to output.
GPIO [2]	J4	DIOH	Serial Flash (SI) Serial Flash Data Input (SI) of Serial Flash. This pin is by default set to output.
GPIO [3]	M1	DIOH	Serial Flash (CE0#) This pin functions as Chip Enable (CE#0) of Serial Flash.
GPIO [4]	M2	DIOH	GPIO [4] Can be configured by customer firmware.
GPIO [5]	H4	DIOH	GPIO [5] Can be configured by customer firmware.
GPIO [6]	M5	DIOH	GPIO [6] Can be configured by customer firmware.
GPIO [7]	L5	DIOH	GPIO [7] Can be configured by customer firmware.

GPIO [8]	H5	DIOH	GPIO[8] Can be configured by customer firmware.
GPIO [9]	M6	DIOH	GPIO[9] Can be configured by customer firmware.
GPIO [10]	L6	DIOH	GPIO[10] Can be configured by customer firmware.
GPIO [11]	K6	DIOH	GPIO[11] Can be configured by customer firmware.
GPIO [12]	J6	DIOH	GPIO[12] Can be configured by customer firmware.
GPIO [13]	H6	DIOH	GPIO [13] Can be configured by customer firmware.
GPIO [14]	M7	DIOH	GPIO [14] Can be configured by customer firmware.
GPIO [15]	L7	DIOH	GPIO [15] Can be configured by customer firmware.
GPIO [16]	K7	DIOH	GPIO [16] Can be configured by customer firmware.
GPIO [17]	J7	DIOH	GPIO [17] Can be configured by customer firmware.
GPIO [18]	H7	DIOH	GPIO [18] Can be configured by customer firmware.
GPIO [19]	M8	DIOH	GPIO [19] Can be configured by customer firmware.
GPIO [20]	L8	DIOH	GPIO [20] Can be configured by customer firmware.
GPIO [21]	H8	DIOH	GPIO [21] Can be configured by customer firmware.
GPIO [22]	M9	DIOH	GPIO [22] Can be configured by customer firmware.
GPIO [23]	L9	DIOH	GPIO [23] Can be configured by customer firmware.
GPIO [24]	K9	DIOH	GPIO [24] Can be configured by customer firmware.
GPIO [25]	J9	DIOH	GPIO [25] Can be configured by customer firmware.
GPIO [26]	H9	DIOH	GPIO [26] Can be configured by customer firmware.

GPIO [27]	G9	DIOH	GPIO [27] Can be configured by customer firmware.
GPIO [28]	M4	DIOH	GPIO [28] Can be configured by customer firmware.
GPIO [29]	H3	DIOH	GPIO [29] Can be configured by customer firmware.
GPIO [30]	K3	DIOH	GPIO [30] Can be configured by customer firmware.
GPIO [31]	M3	DIOH	GPIO [31] Can be configured by customer firmware.

Power Supply Interface

Table 9 Power Supply Interface

Signal Name	Pin Number	Type	Description
VCCO	F4, K5, K8,	PI	3.3V I/O Power Supply
VCCK	C7, D4, D8, E4, E9, G5, G6, G7, G8,	PI	1.0V core Power Supply
SD_AVDDL, P0_AVDDL, AVDDL_PLL, P1_AVDDL, U_AVDDL, S_AVDDL	C3, C4, C6, C8, C10, C11, E3, E10, F10, G3, H10, J10, K1, K2	PI	Analog 1.0V Power Supply
SD_AVDD33, U_AVDDH	D1, K11	PI	Analog 3.3V Power Supply
X_AVDDH	D9	PI	3.3V Crystal Power Supply
AVDD25	C9	PO	Analog 2.5V Regulator Output

5.4 Function Description

5.4.1 LED Indicator

By default, GPIO[13], [14], [15], [17], [19], [27], [28], and [31] are used as LED indicator. They can be configured to support a hardware PWM. If the user has a different application for LED function, please contact JMicon's AE before PCB layout.

5.4.2 GPIO Initial Value

All GPIOs set as input mode and enable internal pull-up function while in reset. After reset, the firmware will program all GPIOs as input mode. Afterward, the initial value of GPIOs is read and stored in the system RAM for future using.

5.4.3 VBUS sense

VBUS detector is used for V_{BUS} detection. There is a voltage divider circuit on a board. Power source is connected from V_{BUS}5V. The output is connected to VBUS_SENSE pin for V_{BUS} detection.

5.4.4 I2C Master

Here are two I2C master interfaces. GPIO[7] and GPIO[11] can be configured one I2C master interface. GPIO[23] and GPIO[24] can be configured the other I2C master interface.

6 Electrical Specifications

6.1 Absolute Maximum Ratings

Warning: Absolute maximum rating may cause the device permanent damage or reliability will be affected. All voltage is a specified reference to the ground unless otherwise specified.

Table 10 Absolute Maximum Ratings

Parameter	Symbol	Min	Typical	Max	Units
Digital 3.3V	VCCO _(ABS)	-0.3	-	3.6	V
Digital 1.0V	VCCK _(ABS)	-0.3	-	1.1	V
3.3V Crystal Pad power	XAVDDH _(ABS)	-0.3	-	3.6	V
Analog 1.0V	AVDDL _(ABS)	-0.3	-	1.1	V
USB VBUS	VBUS	-0.3	-	5.5	V
Digital I/O input voltage	V _{I(D)}	-0.3	-	3.6	V
Storage temperature	T _{STORAGE}	-40	-	150	°C

6.2 Recommended Operating Voltage and Temperature

Table 11 Operating Voltage and Temperature

Parameter	Symbol	Min	Typical	Max	Units
Digital 3.3V Power Supply	VCCO	3.0	3.3	3.6	V
Digital 1.0V Power Supply	VCCK	1.0	1.05	1.1	V
3.3V Crystal Pad Power	XAVDDH	3.0	3.3	3.6	V
Analog 1.0V Power Supply	AVDDL	1.0	1.05	1.1	V

USB VBUS	VBUS	4.5	5.0	5.5	V
Digital I/O input voltage	$V_{I(D)}$	3.0	3.3	3.6	V
Ambient Operation Temperature	T_A	0	-	70	°C
Maximum Junction Temperature	T_j	-	-	125	°C

Note:

1. All supply mean voltage power noise $\leq \pm 5\%$

6.3 External Clock Source Conditions

Table 12 External Clock Source Conditions

Parameter	Symbol	Min	Typical	Max	Units
External reference clock		-30ppm	25	+30ppm	MHz
Clock Duty Cycle		45	50	55	%

6.4 DC Electrical Characteristics

Table 13 DC Specifications

Parameter	Symbol	Min	Typical	Max	Units
Input low voltage	V_{IL}	-0.3	-	0.8	V
Input high voltage	V_{IH}	2	-	5.5	V
Output low voltage	V_{OL}	-	-	0.4	V
Output high voltage	V_{OH}	2.4	-	-	V
Low Level Output Current	I_{OL}	9.7	-	21.5	mA

High Level Output Current	I _{OH}	17.0	-	56.5	mA
Internal weak pull-high resistance	R _{pu}	27	38	59	KΩ

Note:

The above test results are under the environment at ambient temperature 25 °C

6.5 Crystal input

Signal crystal input (25MHz) is needed.

Table 14 Crystal Electrical Specification

Parameter	Symbol	Min	Typical	Max	Units
Crystal start up time vs. AVDDL	TCrystal	-	-	5	mS
Crystal Frequency	fclk	-	25	-	MHz
Long term stability (Crystal Only)	ΔfMAX_Crystal	-30	-	30	ppm
Long term stability (On Board)	ΔfMAX_OnBoard	-150	-	150	ppm
Equivalent Series Resistance	ESR	-	-	55	Ω

6.6 Power Consumption

The power consumption is tested by the following setting environment and IC condition.

➤ Setting Environment

- Operation System: Windows 10 64bit
- Test tool: HD Speed
- Connected device: SATA SSD Samsung 860 Pro / NVMe SSD Samsung 970 Pro

➤ IC Condition

- Device Under Test: Evaluation Board (JMS581-EV-02-01)
- Voltage: 1.05V / 3.3V
- Firmware version: Standard Firmware
- Device: JMS581DL

6.6.1 USB 2.0 to PCIe and SATA

Operation with PCIe L0 state and SATA 6Gb/s

Table 15 Power Dissipation – USB 2.0 to PCIe L0 and SATA 6Gb/s

Parameter	Symbol	Min	Typical	Max	Units	Details/ Conditions
Digital 3.3V	VCCO	-	1.9	-	mA	Operate @ 3.3V Temperature = 25°C
Digital 1.0V	VCCK	-	299.8	-	mA	Operate @ 1.05V Temperature = 25°C
Analog 3.3V	XAVDDH	-	2	-	mA	Operate @ 3.3V Temperature = 25°C
Analog 1.0V	AVDDL	-	598.1	-	mA	Operate @ 1.05V Temperature = 25°C

6.6.2 USB 3.2 Gen1 to PCIe and SATA

Operation with PCIe L0 state and SATA 6Gb/s

Table 16 Power Dissipation – USB 3.2 Gen1 to PCIe L0 and SATA 6Gb/s

Parameter	Symbol	Min	Typical	Max	Units	Details/ Conditions
Digital 3.3V	VCCO	-	1.9	-	mA	Operate @ 3.3V
Digital 1.0V	VCCK	-	337	-	mA	Operate @ 1.05V
Analog 3.3V	XAVDDH	-	2	-	mA	Operate @ 3.3V
Analog 1.0V	AVDDL	-	658.5	-	mA	Operate @ 1.05V

6.6.3 USB 3.2 Gen2 to PCIe and SATA

Operation with PCIe L0 state and SATA 6Gb/s

Table 17 Power Dissipation – USB 3.2 Gen2 to PCIe L0 and SATA 6Gb/s

Parameter	Symbol	Min	Typical	Max	Units	Details/ Conditions
Digital 3.3V	VCCO	-	1.9	-	mA	Operate @ 3.3V

Parameter	Symbol	Min	Typical	Max	Units	Details/ Conditions
Digital 1.0V	VCCK	-	378.9	-	mA	Operate @ 1.05V
Analog 3.3V	XAVDDH	-	2	-	mA	Operate @ 3.3V
Analog 1.0V	AVDDL	-	705.5	-	mA	Operate @ 1.05V

6.7 Power-on Sequence

The power-on sequence is defined in Figure 6. Designers should follow all the rules for external power designs.

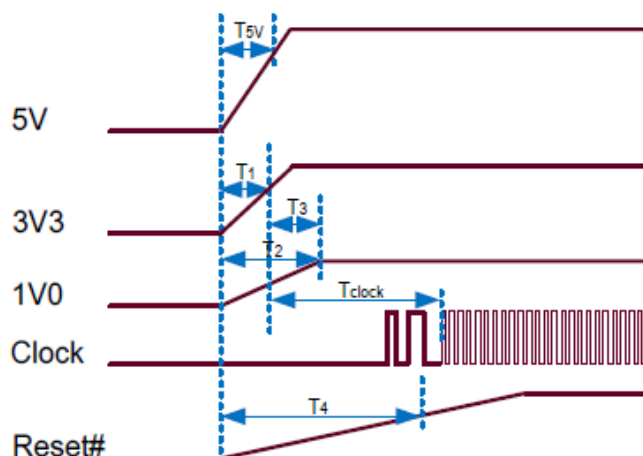


Figure 6 Power-on Sequence

T_{5V}: Rise time for 5V power rail from 10% to 90%

T₁: Rise time for 3V3 power rail from 5V 10% to 3V3 90%

T₂: Rise time for 1V0 power rail from 10% to 90%

T₃: Time interval between 3V3 power and 1V0 Power

T₄: Rise time for RST# signal from 0V to 1V77

T_{clock}: Time interval between 3V3 and 90% clock swing

Note: Clock must meet 25MHz +/-30ppm during the sequence.

The recommended power sequence and timing requirements are listed in Table 25.

Table 18 Power-On Timing Requirements

Time	Minimum	Maximum
T _{5V}	-	20ms
T ₁	0ms	10ms
T ₂	0ms	10ms
T ₃	-5ms	5ms
T ₄	120ms	500ms

Time	Minimum	Maximum
T_{Clock}	-	5ms

The RESET timing constraint is based on the external RC reset circuits. To control the charge and discharge time for RC circuits, minimum, and maximum requirements are listed. If designers apply timing control chip to control the reset signal, the only requirement will be the minimum value. In other words, the maximum value can be ignored without problems.

				Ta: -40 ~ 85°C. J = Copper wired RoHS compliant halogen-free green product; Ta: -40 ~ 85°C.
g (I)	1 digit	Internal bonding type	A, B, C, ...	A, B, C, ...
h (A0)	2 digit	Version of mask ROM	A0, A1, A2, ... B0, B1, B2, ... Z0	Version A0, A1, A2, ... Version B0, B1, B2, ... Version Z0= no mask ROM
i (A)	1 digit	Version of the IC	A, B, C, ...	Version A, B, C, ...

7.3 Top Mark

Each device has its unique top mark containing the information of provider, device name, part number, manufacturing date code, lot number and pin one identifier for identification. The top mark is illustrated in Figure 7 below.

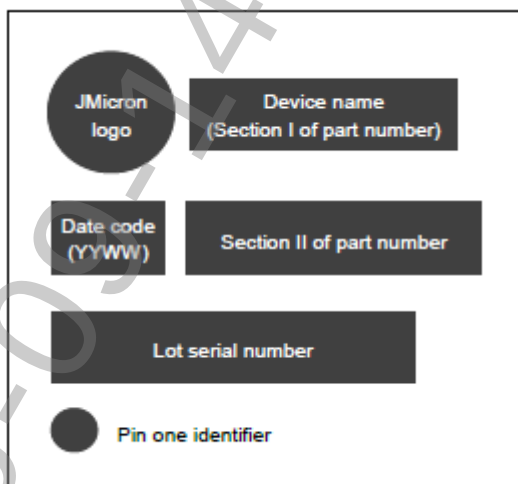


Figure 8 Illustration of Device Top Mark

Note: The above information is only for use in this product datasheet. For more detailed ordering information, please contact to JMicon's Sales Department, representatives, or distributors when ordering parts.

How to Reach Us:

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